Damped two-sphere collision

LAMMPS dem01 | Restitution and automatic normal damping | 28 October 2026

PASS	All four native runs pass the official 2% restitution criterion.

Restitution results

Run	Steps / frames	Measured e	Error to e = 0.5 (%)	Final left vx (m/s)
CPU dt	5100 / 5101	0.4999416729	0.0116654	-0.999883346
GPU dt	5100 / 5101	0.4999416729	0.0116654	-0.999883346
CPU dt/2	10200 / 10201	0.4999827181	0.0034564	-0.999965436
GPU dt/2	10200 / 10201	0.4999827181	0.0034564	-0.999965436

Official saved-state e = 0.499941826448; target e = 0.5. The official check at cumulative step 4100 (41 us) requires relative error below 2%; the separate total x-momentum check also passes. Signed e = (v_right_after - v_left_after) / 4.

At the official time step, maximum checkpoint differences are 4.59e-12 m and 3.07e-07 m/s; 1e-10 pointwise agreement is not claimed. CPU/GPU maximum differences: 0 m and 4.441e-16 m/s. Coarse / half-step absolute e errors: 5.83271e-05 / 1.72819e-05. Refinement approaches target 0.5, not the coarse golden trajectory. Total momentum remains 0 kg m/s.

Inputs and automatic damping

Quantity	Official input / FunDEM configuration
Identical spheres	Diameter 0.005 m; density 2700 kg/m^3; centers x = +/-0.002550 m
Initial motion	vx = +2 / -2 m/s; omega_z = +22 / -22 rad/s
Material / contact	E = 7e10 Pa; Poisson ratio = 0.33; e = 0.5; sliding friction = 0.4
Other effects	Rolling / torsional friction, gravity and global damping = 0
Time / output	dt = 1e-8 s, 5100 steps; half-dt = 5e-9 s, 10200 steps; duration 51 us; every step retained
Damping configuration	Nonlinear Hertz model; explicit damping override off; restitution-based automatic damping active

beta = -ln(e) / sqrt(pi^2 + ln(e)^2) = 0.215453761966
S_n = 2 E* sqrt(R* delta); c_n = 2 sqrt(5/6) beta sqrt(m* S_n).
E* = 39277297721.9 Pa; m* = 8.83572933822e-05 kg. c_n varies with overlap and has units N s/m. No material or damping coefficient was fitted.

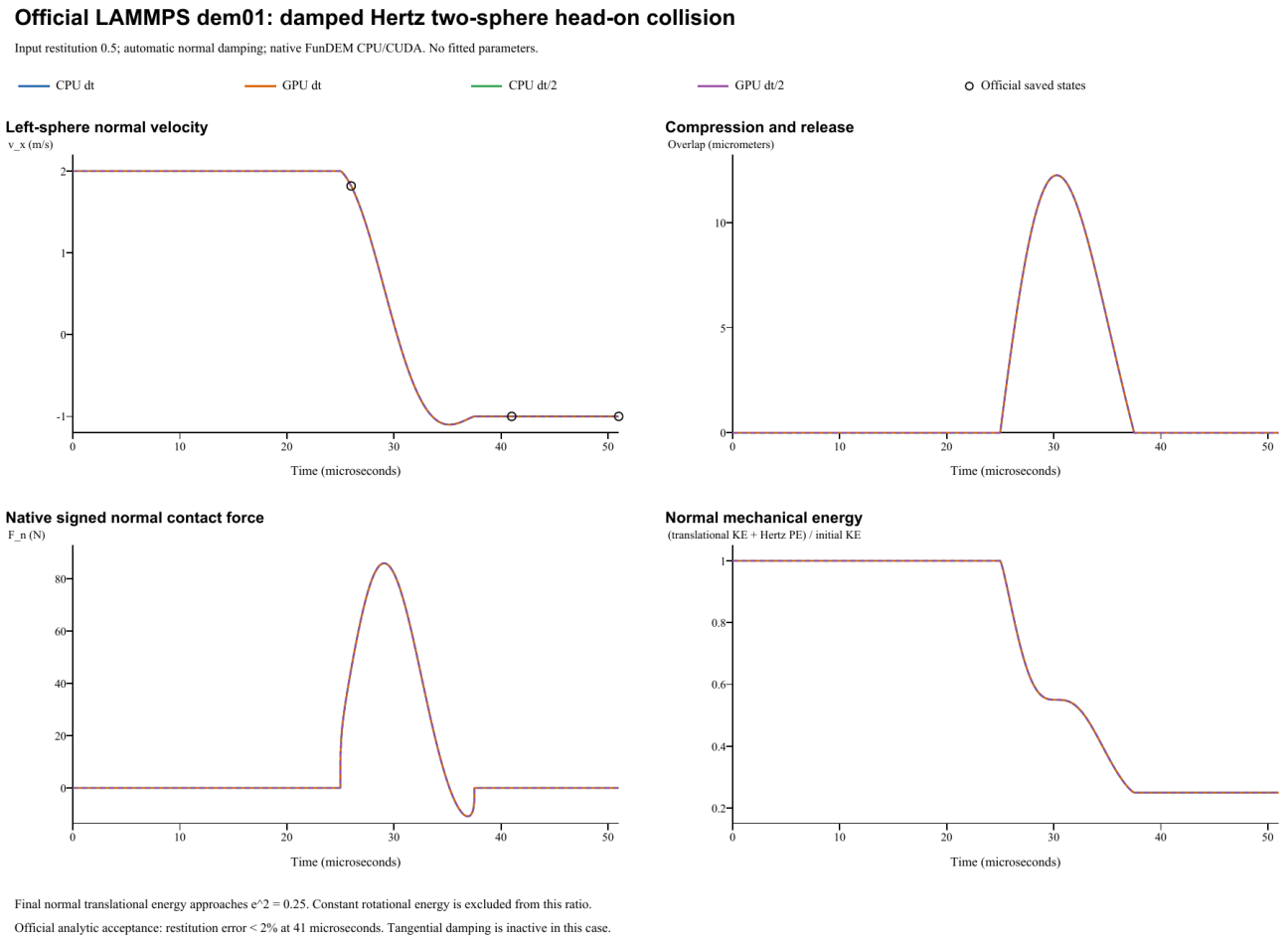
Independent damping audit: reconstruct assembly half-step velocity from completed native velocities and signed native force, then infer c_n = (F_elastic - F_native) / u_half. This is a reconstruction, not direct native c_n output. Samples require delta > 1e-10 m and abs(u_half) > 0.08 m/s. Maximum coefficient error / peak c_n = 2.265e-14; maximum force reconstruction error / peak force = 4.965e-16.

Reference provenance

Official LAMMPS YAML and golden states; commit f9720ce2d176d615495d1d6c31349b1f71be1c73. Local Git blob: 9029229be2a7123a3a5c9aef4d04e2df5a0bc9e0. LAMMPS was not rerun locally. Regression epsilon = 1e-10 is not treated as a cross-code tolerance. FunDEM uses one uninterrupted event; LAMMPS segment-setup force re-evaluations are not mirrored. This protocol difference can affect finite-step damped forces; it is not established as the cause of a measured discrepancy.

Curves and retained evidence

Native states and forces; official saved-state checkpoints



The energy plot includes normal translational kinetic energy and Hertz potential; unchanged rotational energy is excluded. The final kinetic fraction approaches e^2 = 0.25. Weak tensile signed force near release is retained because normal damping is not clipped.

Complete native results are retained

CPU dt: run-20261028-033707-825-d4cbe0d9-71ab-4657-9bfe-a9ea3a79df94

GPU dt: run-20261028-033758-018-5614cf1d-bf48-46b2-9f6d-6b38b20fd470

CPU dt/2: run-20261028-033834-165-61b3bee7-fe6c-4c89-b207-5e5b4b18160c

GPU dt/2: run-20261028-033946-991-7b381bce-5717-4e49-9d37-ae4f1a1219db

Run manifest | Independent analysis JSON | Checkpoint CSV | Contact-model audit. Original projects, run.json, saved frames/contacts, trajectories and native/configuration audits are retained. Per-sample damping reconstruction CSVs are in analysis/dem01/.

Execution and validation limits

Real SimulationSession runs used linked FunDEM 0.4.3 native CPU/CUDA libraries. The helper is not a separate collision integrator; no contact solver code was modified. Byte identity with the installed GUI executable was not established.

Verified scope: normal two-sphere collision, restitution and automatic normal damping. Opposite spins cancel tangential contact velocity; torque is zero. General tangential damping differs (LAMMPS c_t/c_n = 0.5; FunDEM automatic ratio about 0.895765), but is inactive here. This does not validate tangential damping, friction, rolling resistance or many-particle behavior.